STAT 7650. Homework 4 (Due: Tuesday, March 31, 2026)

1. (20 points) Let the training data be $\{(y_i, x_i), y_i \in \{-1, 1\}, x_i \in \mathbb{R}^p, i = 1, \ldots, n\}$. A support vector machine classifies an observation x_0 according to $\text{sign}(\alpha + \beta^T x_0)$, that is, $\hat{y} = 1$ if $\alpha + \beta^T x \geq 0$ and $\hat{y} = -1$ otherwise. The parameters α and β can be estimated by solving the following optimization problem,

$$\min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \sum_{i=1}^{n} (1 - y_i(\alpha + \beta^T x_i))_+ + \frac{1}{2}\lambda \sum_{k=1}^{p} \beta_k^2,$$

where the hinge loss $(u)_+ = u$ if $u > 0$ and $= 0$ otherwise.

   (a) Formulate the optimization problem as a quadratic programming.

   (b) Derive the dual problem of this quadratic programming.

   (c) Write out the KKT conditions corresponding to this optimization problem.

   (d) Each female horseshoe crab has a male crab residing in her nest. Satellite males are additional male crabs that reside nearby. Let the response variable be $Y = 1$ if the female crab has at least one satellite male and $Y = -1$ otherwise. The predictor variable is the width of the female crab. In the data file, `satell` denotes the number of satellite males and `width` denote the female crab's width.

   `http://www.auburn.edu/~zengpen/teaching/STAT-7030/datasets/crab.txt`

   What are the estimates of α and β? Set $\lambda = 0.5$. You may use Gurobi, CVXR, or other packages for quadratic programming, but not packages for SVM.

2. (20 points) A variant of the support vector machine employs an ℓ_1-norm penalty on β and estimate α and β by solving

$$\min_{\alpha \in \mathbb{R}, \beta \in \mathbb{R}^p} \sum_{i=1}^{n} (1 - y_i(\alpha + \beta^T x_i))_+ + \lambda \sum_{k=1}^{p} |\beta_k|.$$

   (a) Formulate the optimization problem as a linear programming.

   (b) Derive the dual problem of this linear programming.

   (c) Write out the KKT conditions corresponding to this optimization problem.

   (d) Fit the crab data. What are the estimates of α and β? Set $\lambda = 0.5$. You may use Gurobi, CVXR, or other packages for linear programming, but not packages for SVM.

3. (20 points) Evaluate the following integral using different methods.

$$I = \int_2^{\infty} \frac{1}{\pi(1 + x^2)} dx.$$

(a) What is the integral using the `integrate()` function?

(b) Draw $n = 10,000$ samples from the Cauchy(0, 1) distribution. Compute the Monte Carlo estimate of the integral and the standard error of the estimate.

(c) Draw $n = 10,000$ samples from the shifted exponential distribution with density $f(x) = e^{-(x-2)}$ for $x > 2$. Compute the Monte Carlo estimate of the integral and the standard error of the estimate.

(d) Draw $n = 10,000$ samples from the uniform(0, 0.5) distribution. Compute the Monte Carlo estimate of the integral and the standard error of the estimator. (Hint: consider the change of variable $y = 1/x$.)

4. (15 points) Derive an algorithm to generate random variables using the inverse transformation method, that is, by inverting the cumulative distribution function. Draw $n = 10,000$ random numbers using your algorithm and draw a histogram to compare the empirical distribution with its true pmf or pdf.

   (a) Geometric distribution with pmf $P(X = k) = (1-p)^k p$ for $k = 0, 1, 2, \ldots$.

   (b) Rayleigh distribution with density $f(x) = xe^{-x^2/2}$ for $-\infty < x < \infty$.

   (c) Normal distribution $N(0, 1)$. Approximate the inverse CDF by linear interpolation at a grid of points $\{-5, -4.5, -4, \ldots, 4.5, 5\}$.

5. (15 points) Let X be a Gamma(r, 1) random variable with $r \geq 1$. Let Y be a random variable related to X via $X = t(Y) = a(1 + bY)^3$, where $a = r - (1/3)$ and $b = 1/\sqrt{9a}$. Draw X via Y using the rejection sampling with the envelope

$$e(y) = \frac{a^{-1/6}}{\Gamma(r)} \exp\left\{-\frac{y^2}{2} + a \log a - a\right\}, \qquad y > -\frac{1}{b}.$$

   (a) Prove that $e(y) \geq f(y)$ and identify the expected acceptance rate.

   (b) Implement the squeezed rejection sampling algorithm with the squeezing function

$$s(y) = \frac{a^{-1/6}}{\Gamma(r)}(1 - 0.0331y^4) \exp\left\{-\frac{y^2}{2} + a \log a - a\right\}, \qquad y > -\frac{1}{b}.$$

   (c) Test your code with $r = 2$ by generating $n = 10,000$ random numbers. Plot a histogram of the simulated sample and compare the empirical distribution with the true distribution. What is the acceptance rate in this case? What is the probability that the squeezing step saves computation?

6. (10 points) Let $0 < a < b$. The truncated standard normal distribution has the density

$$f(x) = \frac{\phi(x)}{\Phi(b) - \Phi(a)}, \quad 0 < a < x < b,$$

where $\phi(\cdot)$ and $\Phi(\cdot)$ are the density and the CDF of $N(0,1)$. The truncated Rayleigh distribution has the density

$$g(x) = \frac{xe^{-x^2/2}}{e^{-a^2/2} - e^{-b^2/2}}, \quad 0 < a < x < b.$$

Derive a rejection sampling algorithm for generating samples from a truncated normal distribution using a truncated Rayleigh distribution as the proposal distribution.

(a) Find a constant M such that $Mg(x) \geq f(x)$ for $0 < a < x < b$.

(b) Derive an appropriate algorithm for sampling from a truncated Rayleigh distribution, and then use rejection sampling with this distribution as the proposal to generate samples from a truncated normal distribution. What is the acceptance rate?